

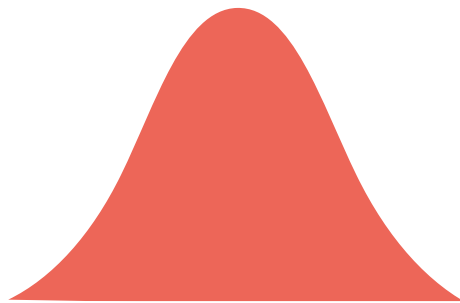
requires equal mass

enforce it “softly”

α



$\int \alpha$

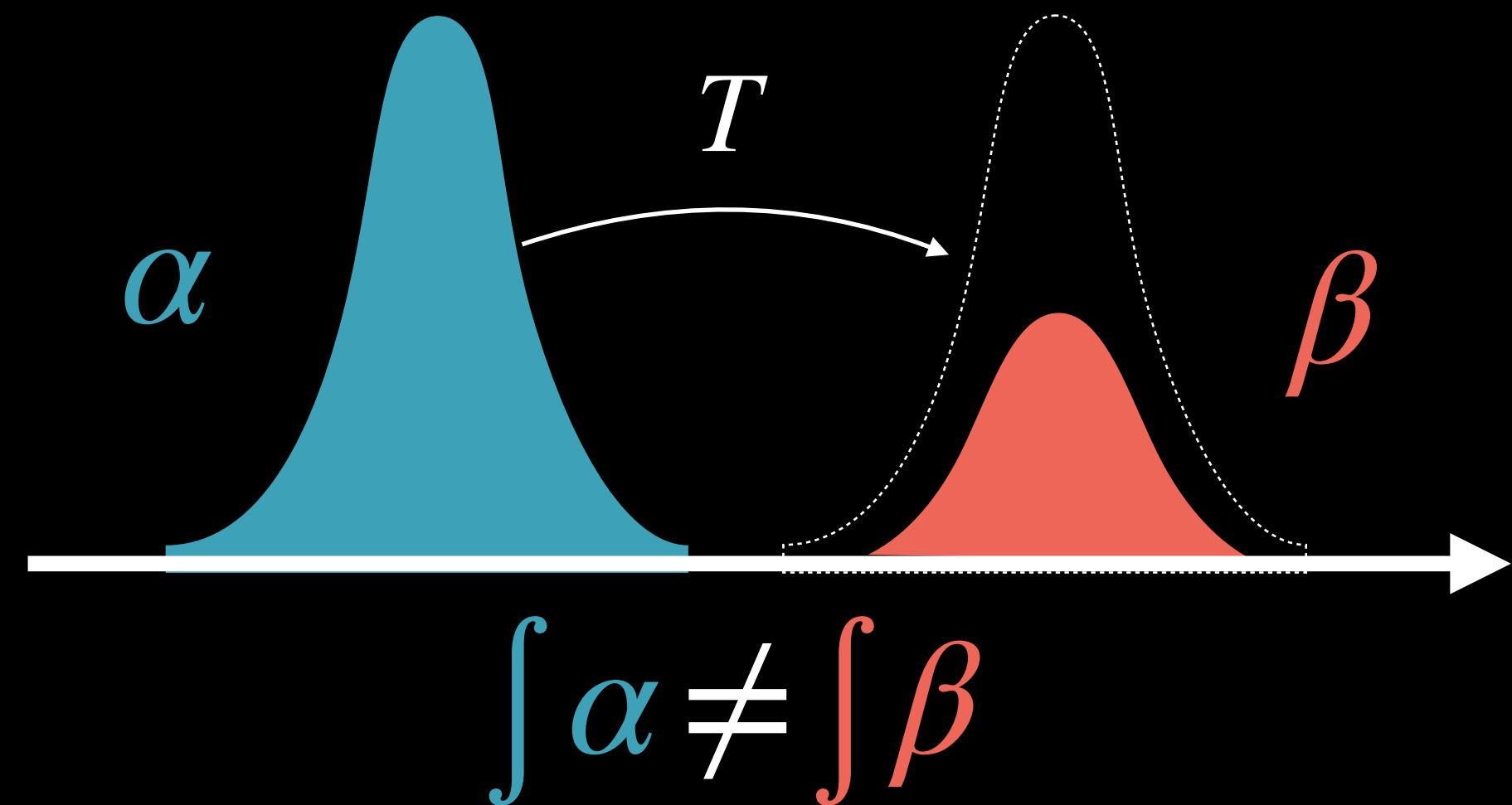


β

$\int \beta$

Unbalanced OT: Motivation

- Classic OT **requires equal total mass** → unrealistic for noisy or partial data.
- Examples:
 - single-cell sequencing data
 - subset selection
 - signal fading, missing measurements
- Unbalanced OT: drop requirement of equal total mass, **enforce it “softly”** through penalties on marginal constraints



Unbalanced OT: Formulation